

Problem 1. Let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be a subset of a vector space V over the real numbers. What does it mean for the set S to be *independent*?

The set S is called independent if for all scalars $a_1, a_2, \dots, a_k \in \mathbb{R}$ we have

$$a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k = \mathbf{0} \implies a_i = 0 \text{ for all } i.$$

Problem 2. Let V be a vector space over the real numbers and let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \subseteq V$ be an independent set of vectors. Consider the span

$$U := \text{Span}(S) = \{a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k : a_1, \dots, a_k \in \mathbb{R}\}.$$

If $\mathbf{v} \in V$ is a vector **not** in U , prove that $\{\mathbf{v}, \mathbf{u}_1, \dots, \mathbf{u}_k\}$ is an independent set.

Proof. Let S be independent and suppose that $\mathbf{v} \notin \text{Span}(S)$. To show that the set $S \cup \{\mathbf{v}\}$ is independent, suppose that we have

$$a\mathbf{v} + a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k = \mathbf{0}$$

for some scalars $a, a_1, a_2, \dots, a_k \in \mathbb{R}$. We need to show that $a = a_1 = \dots = a_k = 0$. If $a \neq 0$ then we would have

$$\mathbf{v} = -\frac{a_1}{a} \mathbf{u}_1 - \frac{a_2}{a} \mathbf{u}_2 - \dots - \frac{a_k}{a} \mathbf{u}_k,$$

which contradicts the fact that $\mathbf{v}$ is not in the span of S . Hence $a = 0$. But then we have

$$a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k = \mathbf{0},$$

which implies that $a_i = 0$ for all i , because S is independent. □

Corollary. (This was not part of the quiz.) Any independent set can be extended to a basis.

Proof. Let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \subseteq V$ be an independent set. If $\text{Span}(S) = V$ then S is a basis. Otherwise, choose any $\mathbf{u}_{k+1} \in V \setminus \text{Span}(S)$. By the previous result, the set $S' := S \cup \{\mathbf{u}_{k+1}\}$ is still independent. If $\text{Span}(S') = V$ then S' is a basis. Otherwise, we continue to create a chain of independent sets $S \subsetneq S' \subsetneq S'' \subsetneq \dots$. If V is finite dimensional then this process must eventually stop,¹ and the final set in the chain is a basis that contains S .

¹This is actually the hard part. It relies on the Steinitz exchange lemma.